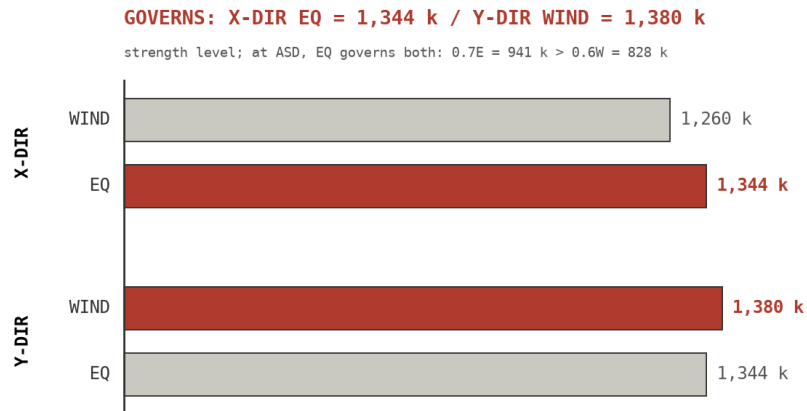


1 GOVERNING LATERAL LOAD

115 GILES WIND vs EQ BASE SHEAR

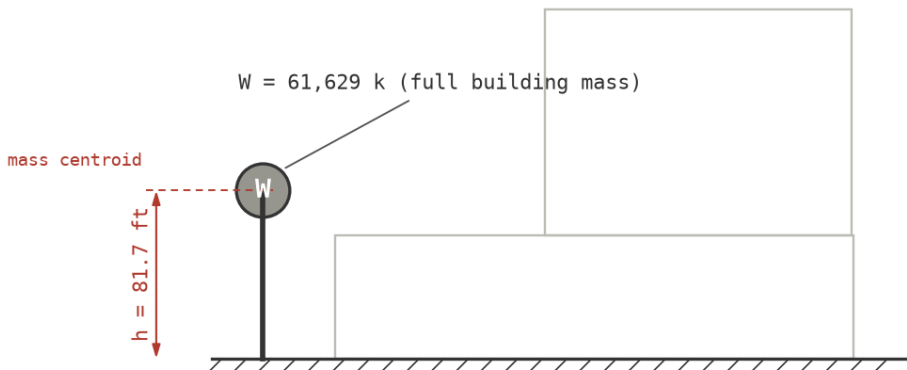


source: 115-GILES-PRELIM-LATERAL-LOADING (ASCE 7-16 case)

2 SDOF IDEALIZATION

115 GILES SDOF LOLLIPOP IDEALIZATION

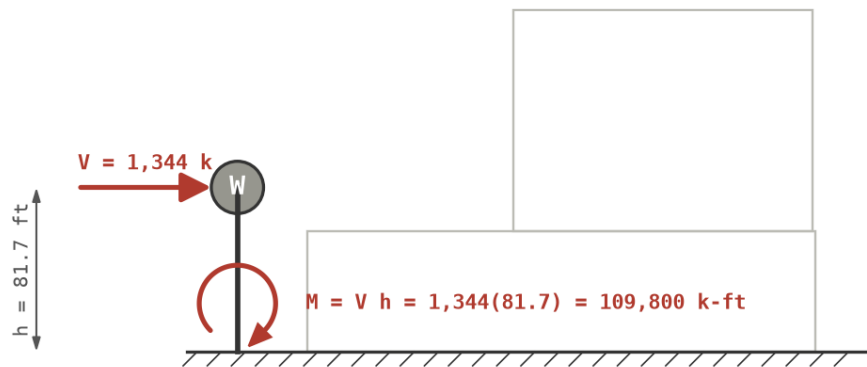
whole-building mass lumped at mass centroid height



3 EQ OVERTURNING MOMENT

115 GILES EQ OVERTURNING MOMENT

EQ base shear applied at mass centroid of SDOF lollipop



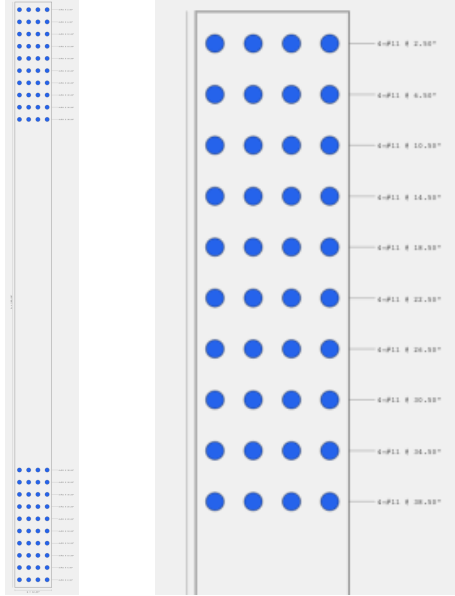
4 WALL QUANTITY

5 walls at 16 ft in X, 3 walls at 16 ft in Y. Each straight wall taken separately; torsion neglected. Demand split evenly; governing case per direction (X: EQ, Y: wind, resultant at $h/2 = 85$ ft).

$$\text{X: } V_u = \frac{1344}{5} = 269 \text{ k} \quad M_u = \frac{109,800}{5} = 21,960 \text{ k-ft} \quad (1)$$

$$\text{Y: } V_u = \frac{1380}{3} = 460 \text{ k} \quad M_u = \frac{1380(85)}{3} = 39,100 \text{ k-ft} \quad (2)$$

Section: 12 in $\times$ 16 ft, $f'_c = 5$ ksi, $f_y = 60$ ksi. Each end: 10 rows of 4#11 at 4 in. Section and flexural capacity from doubly-reinforced section analysis:



$$\begin{aligned} A_s \text{ (each end)} &= 62.46 \text{ in}^2 \\ d / d' &= 171.5 / 20.5 \text{ in} \\ a / c &= 28.4 / 35.5 \text{ in} \\ \varepsilon_t &= 0.0115 \text{ (tension-controlled)} \\ \phi M_n &= 43,124 \text{ k-ft} \end{aligned}$$

$$\phi V_n = \phi A_{cv} (2\lambda \sqrt{f'_c} + \rho_t f_y) = 0.75(2304)(141 + 150)/1000 = 504 \text{ k}$$

$$\begin{aligned} \varepsilon_t &= \text{net tensile strain, extreme tension steel} \\ A_{cv} &= \text{wall shear area, } 12 \times 192 = 2,304 \text{ in}^2 \\ \rho_t &= \text{horizontal web reinf ratio, } 0.0025 \text{ min assumed} \end{aligned}$$

$P = 0$ and equal end reinforcement; wall treated as vertical doubly-reinforced beam.

full wall / end zone

WALL	DCR V	DCR M
X (5 at 16 ft)	$269/504 = 0.53$	$21,960/43,124 = 0.51$
Y (3 at 16 ft)	$460/504 = 0.91$	$39,100/43,124 = 0.91$

Quantity works. Y walls at DCR ≈ 0.9 – tight; X has margin.